Exercise 2 – TechnotzDS

Face Identification

Background

You are Mossad agents operating in Iran. After years of effort, you have successfully hacked the security cameras at the Natanz nuclear facility. Now that you have successfully breached the system, you face one final challenge: automatically identifying the individuals captured on camera to be alerted whenever a person of interest is spotted among the regular workers

You are given a labeled dataset of grayscale face images taken from security cameras.
Each image belongs to one of 28 individuals, and your task is to identify the person appearing in a new unseen image.

The Data

You are given a dataset of **15,988 labeled grayscale images** of **28 different people**.
Each image has resolution **64 × 72** pixels.

Pixel values are grayscale intensities in the range **[0, 255]**, where:

- 0 = black
- 255 = white

File names are of the form:

`pi_j.pgm`

where:

- `i` is the identity label, from `0` to `27`
- `j` is the image index for that person

Main Objective

Build a learning pipeline that predicts the correct identity for each unseen image.

you are expected to think carefully about:

- data representation
- preprocessing choices
- model selection
- regularization
- and generalization ability

Your solution should not rely on trying a single model only.
Instead, you should compare several candidate approaches and justify your final choice both **empirically** and **theoretically**.

Part A – Data Representation and Preprocessing

Before training any model, examine the dataset carefully and decide how the images should best be represented for the classification task.

You should consider questions such as:

- Is any preprocessing needed?
- Should you work directly with the raw pixels?
- Are there structures or artificial changes in the data that may interfere with learning?
- Can you construct a representation that helps the model generalize better?

In your report, clearly describe:

1. What preprocessing steps you chose
2. Why you chose them
3. How you expect them to affect the quality of the classification

Part B – Supervised Models: Introduction and Motivation

Introduce and justify at least three supervised models.

You may choose your models from families such as:

- linear models,
- margin-based models,
- instance-based models ,
- tree-based models,
- or ensemble methods.

You do **not** have to use all of these, but you must compare at least **three** substantially different models. **At least two of the three models must be implemented independently by you, without using external libraries that already solve the learning algorithm itself.**

For each model:

1. Explain briefly how it works

2. State which hyperparameters you tuned and explain how you performed the tuning.
3. Explain what type of regularization or complexity control it uses
4. Briefly describe the bias/variance behavior you would expect from this model. A more detailed discussion will be provided in the next section.
5. Explain why this model is a reasonable candidate for the task

At this stage, your goal is to introduce the models and build intuition about their expected behavior.
The actual empirical comparison between models will be carried out later using K-Fold Cross Validation.

Part C – Regularization and Model Complexity

For each model you choose, investigate how controlling model complexity affects performance.

In your analysis, explain:

- which settings make the model simpler or more flexible,
- how these choices relate to underfitting versus overfitting,
- and how regularization helps improve generalization.

For each model, you should go beyond naming the hyperparameters and discuss them conceptually. In particular, think carefully about what makes this specific model complex, what kind of regularization or complexity control is appropriate for it, and why. Explain how the relevant mechanism works: what it encourages the model to do, what it prevents, and what kind of behavior you would expect when its strength is too weak or too strong.

Your discussion should address questions such as:

- What makes this model more constrained or more expressive?
- Which hyperparameters directly control that tradeoff?
- What kind of regularization is natural for this model, and why does it fit this model better than other forms of regularization?

Part D – Model Selection with Nested K-Fold Cross Validation

In this part, use **nested K-Fold cross validation** to perform both **hyperparameter tuning** and **model selection** in a principled and fair way.

Requirements

- Use an **outer K-Fold loop** for the final evaluation and comparison of the candidate models.
- In each outer split, treat one fold as an **outer test fold** and use the remaining folds as the **outer training set**.
- For each candidate model, use an **inner K-Fold loop** on the outer training set in order to select hyperparameters.
- Apply the same outer and inner cross-validation protocol to all candidate models, so that comparisons are fair.
- At the end of the process, select the model that achieves the best **average outer-fold performance**.

Procedure

- For each outer fold:
- Split the data into:
- an **outer training set**
- and an **outer test fold**
- For each candidate model:
- run **inner K-Fold cross validation** using only the **outer training set**
- evaluate several hyperparameter settings
- select the hyperparameter setting that achieves the best **average inner validation score**
- Retrain the model on the **full outer training set** using the hyperparameters selected in the inner loop.
- Evaluate the retrained model on the **outer test fold**.
- Store, for this outer fold:
- the test score of each model
- and the hyperparameters selected for each model
- Repeat this process for all outer folds.

After completing all outer folds

For each model:

- compute the **average outer test score**
- examine the **variability across outer folds**
- summarize which hyperparameter settings were selected across the different outer folds

Finally:

- compare the candidate models based on their **outer cross-validation performance**
- choose the final selected model accordingly

After selecting the final model based on the nested cross-validation procedure, retrain it on the full labeled training set using the selected hyperparameters, and use it to predict the unlabeled test set.

Theoretical discussion

In your report, address the following:

- Why is **nested** cross validation preferable here over relying on a single train/validation split?
- Why is it important to separate **hyperparameter tuning** from **final model evaluation**?
- What is the role of the **inner loop**, and what is the role of the **outer loop**?
- How does the choice of K affect the validation process? In particular: How does using a small value of K affect the size of the training set in each fold?

Part E [Bonus- 15 p] – Exploring Unsupervised Learning on This Task

As an optional exploratory exercise, you may apply one simple unsupervised method to this problem and analyze its behavior.

The purpose of this bonus part is **not** to obtain strong classification performance, but rather to explore whether simple unsupervised methods can discover any meaningful structure in the data, and to understand their limitations in this setting.

Possible methods include:

- **K-Means**
- **Gaussian Mixture Model (GMM)**

You may apply the method either to the raw images or to a simple feature representation derived from them.

If you choose to complete this bonus part, address the following:

- Which method did you try?
- How did you represent the images before applying the method?
- To what extent did the resulting groups align with the true identities?
- What kinds of visual patterns did the method seem to capture?
- Why are these patterns insufficient, or only partially sufficient, for the true task?

- What does this reveal about the limitations of simple unsupervised methods on this dataset?

Part F – Theoretical Discussion: Bias–Variance Tradeoff

Your submission must include a substantial theoretical discussion of the **bias–variance tradeoff**.

Address the following:

E1. General explanation

Explain in general:

- what high bias means,
- what high variance means,
- how these phenomena appear in classification problems,
- and why balancing them is important.

E2. In the context of this exercise

Discuss how the structure of this dataset and the synthetic perturbation we added to the images may affect bias and variance.

E3. Per-model discussion

For each of the models you tested, explain:

- whether it is more prone to bias or variance,
- how its hyperparameters affect that tradeoff,
- and how regularization changes its behavior.

What to Include in Your Report

Your report should include a clear summary of your experimental process and results.

In particular, you should present:

1. The pipeline you used
2. The candidate models
3. The hyperparameters examined
4. The cross-validation results

5. The final selected model
6. A discussion of the results

You are encouraged to include:

- a comparison table,
- plots if relevant,
- discussion of stability across folds,
- and examples of common failure cases if you notice any.

Submission Requirements

You need to submit:

1. Code

A python file named:

`ex2_name_id.py`

This file should include your full solution:

- data loading
- preprocessing
- model training
- cross-validation
- hyperparameter tuning
- final prediction on the test set

2. Predictions

A file named:

`results_name_id.csv`

containing the predicted labels for the unlabeled test images.

3. Report

A PDF named:

`ex2_name_id.pdf`

Your report must include:

- preprocessing choices
- comparison of multiple models

- regularization and complexity analysis
- K-Fold Cross Validation results
- theoretical discussion of bias–variance
- final conclusions